INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATIONS

Mid-Term Examination, Spring Semester, Date: 10 March 2025

Time Allowed: 1.5 hours **Total Marks: 30**

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1. Assume that the input message sequence is $\{1.2, -0.2, -0.5, 0.4, 0.89, 1.3\}$. Determine the **quantized sequence** and the corresponding **binary bitstream**, if the input sequence is quantized in the range (-1.5, 1.5) with 4 levels using a

(a) uniform quantizer (2 marks)

(b) μ-law quantizer where $\mu = 9$ (3 marks)

* **NOTE:** A μ-law compressor with normalized input x and normalized output y is defined by

$$y = \pm \frac{ln(1 + \mu|x|)}{ln(1 + \mu)}$$

2. A message signal $m(t) = (4/\pi)\sin(2\pi 500t)$ is sampled, and delta modulation is used for digital representation and transmission. In order to meet the requirements that there be no slope overload noise and that the granular noise can not exceed 0.1 volts,

(a) determine the maximum step size (1 mark)

(b) determine the minimum sampling frequency (2 marks)

3. Increase in signal-to-quantization noise ratio with bit rate is much more dramatic for pulse code modulation (PCM) than that for delta modulation (DM). Justify this statement using an example. (3 marks)

4. Prove mathematically that the minimum mean square error achieved by a linear optimum predictor (Wiener filter) is always less than the variance of the input signal that is being predicted. (3 marks)

5. Given a multiplexer with sampling frequency of 8 KHz that accommodates six input signals having bandwidths of 8 KHz, 3.5 KHz, 2 KHz, 1.8 KHz, 1.5 KHz and 1.2 KHz. **In the multiplexer output, it is ensured that successive samples of each input signal are equispaced in time.**

(a) Design the multiplexer involving submultiplexers and frame marker. (4 marks)

(b) Determine the resulting bandwidth of the multiplexer. (1 mark)

6. A band-limited signal $y(t)$ is impulse sampled with a sampling time period of T_s. If $Y(\omega) = 0$ for $|\omega| \geq \frac{\pi}{T_s}$, determine a reconstruction filter $h(t)$ such that (2 marks)

$$\frac{dy(t)}{d(t)} = \sum_{n=-\infty}^{\infty} y(nT_s)h(t - nT_s)$$

7. Consider a signal s(t) as shown in Figure 1. A filter $h(t)$ is designed by matching it to s(t). With s(t) as input, the output of $h(t)$ is sampled at 800 Hz.

0.5

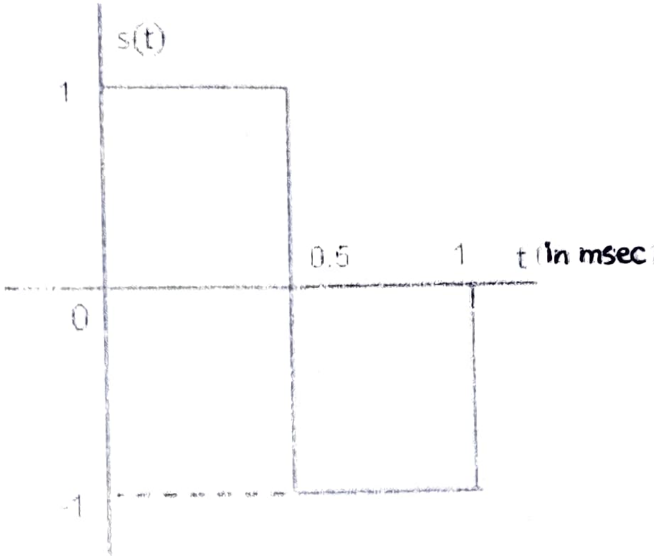

Figure 1

(a) Determine $h(t)$. (1 mark)

(b) Sketch $h(t)$. (1 mark)

(c) Determine the output of $h(t)$. (2 marks)

(d) Sketch the output of $h(t)$. (1 mark)

8. (a) Define the frequency-domain Nyquist criterion for zero inter-symbol interference. (1 mark)

(b) Consider a pulse of duration T_b whose spectrum is shown in Figure 2. Justify with explanation whether this spectrum will satisfy the Nyquist criterion for zero inter-symbol interference for

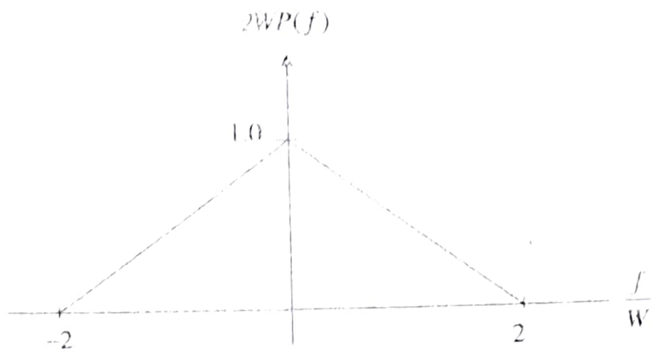

Figure 2

(i) $T_b < 1/(2W)$ (1 mark)

(ii) $T_b = 1/(2W)$ (1 mark)

(iii) $T_b > 1/(2W)$ (1 mark)

NOTE: Answers without justification will not be considered

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End-Term Examination, Spring Semester, 2025

Time Allowed: 3 hours **Total Marks: 40**

NOTE: Attempt ALL questions. All parts of a question must be answered in continuation. Make justifiable assumptions, if necessary, and clearly state these.

1. An M-ary PAM system transmits binary data at a rate of 4×10^6 bits/sec over a bandlimited AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz using the raised cosine roll-off characteristic shown in the Figure 1.

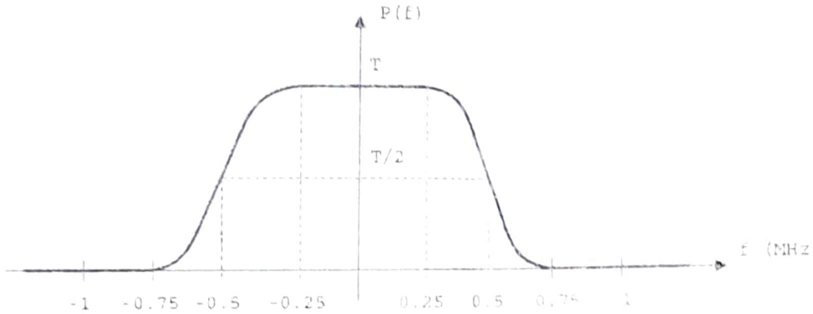

Figure 1

(a) Determine the roll-off factor and the value of M for the PAM system. (2 marks)

(b) Determine the minimum average bit energy-to-noise power spectral density ratio E_b/N_0 (in dB) required to achieve a bit error probability less than 10^{-3}. (2 marks)

2. The overall pulse response $P(f)$ of a duobinary signalling scheme is generated by the block shown in Figure 2, where $H_N(f)$ is an ideal Nyquist (low-pass) filter with cut-off frequency of $W = 1/(2T)$ Hz.

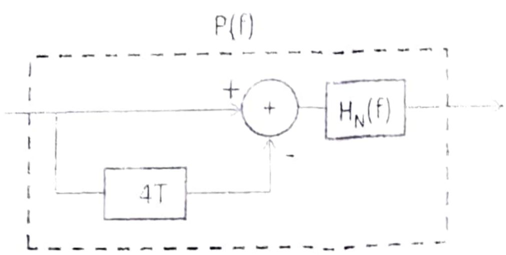

Figure 2

(a) Determine $P(f)$, and sketch the magnitude and phase response. (3 marks)

(b) Determine the time-domain pulse $p(t)$ and sketch it. (2 marks)

(c) Using $P(f)$ from part (a), determine the frequency response of the corresponding transmit and receive matched filters. (2 marks)

3. Consider a two level PAM system with transmitted levels 1 and -1. The channel through which the data is transmitted introduces intersymbol interference and corresponding received signal is given by

$$y_k = I_k + 0.2I_{k-1} + n_k$$

where y_k, I_k and n_k represent the received sample, input sample, and AWGN sample at k^{th} time instant, respectively. The received samples at $k = 1$ to 4 are $\{0.8, -0.2, 0.4, 1\}$. Use the maximum likelihood sequence estimation based on Viterbi algorithm to determine the surviving sequence. (4 marks)

4. A binary communication system uses the following pair of waveforms for binary signaling

$$s_1(t) = u_T(t)$$
$$s_2(t) = \cos(2\pi t/T)\ u_T(t)$$

where $u_T(t) = u(t) - u(t-T)$, and $u(t)$ is the unit step function.

(a) Find a complete set of basis functions to represent these two signals. (2 marks)

(b) Find the signal vectors $\mathbf{s}_1$ and $\mathbf{s}_2$, and plot them in the signal space. (1 mark)

(c) Assume that $s_1(t)$ and $s_2(t)$ are used with equal probability in an AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz. Find the probability of bit error in terms of average bit energy-to-noise PSD ratio E_b/N_0. (2 marks)

5. Consider the constellation diagram shown in Figure 3 used for data transmission over an AWGN channel with two-sided noise power spectral density $N_0/2$ watts/Hz. Assume that the symbols are grey encoded.

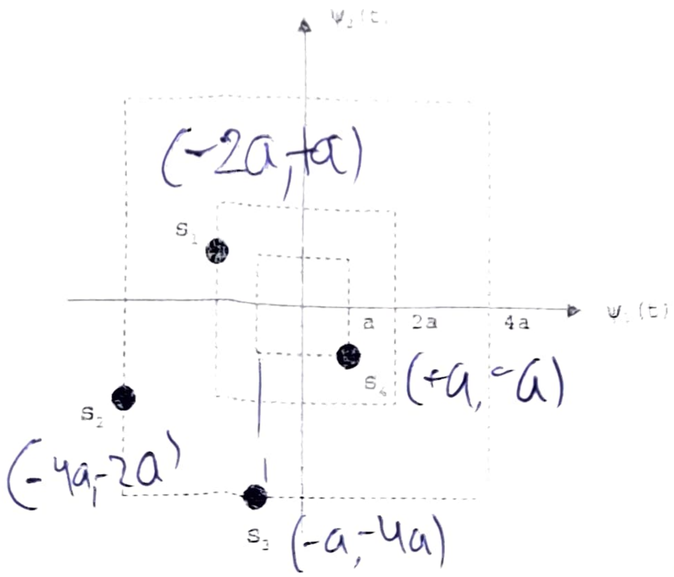

Figure 3

(a) Find the probability of bit error in terms of average bit energy. (2 marks)

Compare the result in (a) with the corresponding expression for a conventional QPSK system, and determine the loss or gain in SNR in dB if the probability of bit error is kept same for both the systems. (2 marks)

6. (a) Justify mathematically, why the minimum separation between the two frequencies used in binary FSK for ensuring orthogonality does not guarantee continuous phase at bit transitions. (2 marks)

   (b) Define the expression for a general continuous phase FSK signal. (1 mark)

   (c) Justify mathematically, how minimum shift keying (MSK) maintains minimum separation between two frequencies in addition to continuous phase at bit transitions. (2 marks)

   (d) Draw the phase state trellis for the MSK signals for the time duration $[0, 3T]$ with initial phase of $\pi/2$. (1 mark)

7. Assume that X is a uniformly distributed random variable in the interval $[\theta\ 1]$ whose probability density function is defined as

$$f_\theta(x) = \begin{cases} \dfrac{1}{1-\theta} & ; \theta \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

The samples $\{x_1, \ldots, x_N\}$ were drawn independently from this distribution. Find the maximum likelihood estimate of θ. (2 marks)

8. Assume a mid-rise type symmetric uniform quantizer having L number of quantization levels.

   (a) Derive the output signal-to-noise ratio if the input X is uniformly distributed over $[-0.5, 0.5)$ and quantization error is uniformly distributed over $[-\Delta/2, \Delta/2)$ where Δ represents the step size of the quantizer. (1 mark)

   (b) Suppose the quantizer in (a) is changed to another quantizer by uniformly distributing X over $[-1, 1)$. The characteristic $f(X)$ of the new quantizer is defined as

$$f(X) = \begin{cases} -0.5 & ; \text{for } -1 \leq X < -0.5 \\ \text{Uniform Quantizer with step size } \Delta = 1/L & ; \text{for } -0.5 \leq X < 0.5 \\ 0.5 & ; \text{for } \ 0.5 \leq X < 1 \end{cases}$$

   Determine the variance of the quantization error. (3 marks)

   (c) What will be the resulting output SNR for the new quantizer in (b)? (1 mark)

   (d) Is it possible to increase SNR beyond the value obtained in (c) by increasing the number of bits required per sample? Justify your answer. (1 mark)

9. Consider the continuous signal

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)$$

   (a) Assume that an ideal low pass filter is used for reconstruction process. Is it possible to reconstruct $x(t)$ exactly if $x(t)$ is sampled at the Nyquist rate? Justify your answer with sufficient reasoning. (1 mark)

   (b) Assuming that the sampling rate is fixed at the Nyquist rate due to hardware limitation, provide a solution for exact reconstruction of $x(t)$. (1 mark)

Complementary Error Function Table

x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)	x	erfc(x)
0	1.000000	0.5	0.479500	1	0.157299	1.5	0.033895	2	0.004678	2.5	0.000407	3	0.00002209
0.01	0.988717	0.51	0.470756	1.01	0.153190	1.51	0.032723	2.01	0.004475	2.51	0.000386	3.01	0.00002074
0.02	0.977435	0.52	0.462101	1.02	0.149162	1.52	0.031587	2.02	0.004281	2.52	0.000365	3.02	0.00001947
0.03	0.966159	0.53	0.453536	1.03	0.145216	1.53	0.030484	2.03	0.004094	2.53	0.000346	3.03	0.00001827
0.04	0.954889	0.54	0.445061	1.04	0.141350	1.54	0.029414	2.04	0.003914	2.54	0.000328	3.04	0.00001714
0.05	0.943628	0.55	0.436677	1.05	0.137564	1.55	0.028377	2.05	0.003742	2.55	0.000311	3.05	0.00001608
0.06	0.932378	0.56	0.428384	1.06	0.133856	1.56	0.027372	2.06	0.003577	2.56	0.000294	3.06	0.00001508
0.07	0.921142	0.57	0.420184	1.07	0.130227	1.57	0.026397	2.07	0.003418	2.57	0.000278	3.07	0.00001414
0.08	0.909922	0.58	0.412077	1.08	0.126674	1.58	0.025453	2.08	0.003266	2.58	0.000264	3.08	0.00001326
0.09	0.898719	0.59	0.404064	1.09	0.123197	1.59	0.024538	2.09	0.003120	2.59	0.000249	3.09	0.00001243
0.1	0.887537	0.6	0.396144	1.1	0.119795	1.6	0.023652	2.1	0.002979	2.6	0.000236	3.1	0.00001165
0.11	0.876377	0.61	0.388319	1.11	0.116467	1.61	0.022793	2.11	0.002845	2.61	0.000223	3.11	0.00001092
0.12	0.865242	0.62	0.380589	1.12	0.113212	1.62	0.021962	2.12	0.002716	2.62	0.000211	3.12	0.00001023
0.13	0.854133	0.63	0.372954	1.13	0.110029	1.63	0.021157	2.13	0.002593	2.63	0.000200	3.13	0.00000958
0.14	0.843053	0.64	0.365414	1.14	0.106918	1.64	0.020378	2.14	0.002475	2.64	0.000189	3.14	0.00000897
0.15	0.832004	0.65	0.357971	1.15	0.103876	1.65	0.019624	2.15	0.002361	2.65	0.000178	3.15	0.00000840
0.16	0.820988	0.66	0.350623	1.16	0.100904	1.66	0.018895	2.16	0.002253	2.66	0.000169	3.16	0.00000786
0.17	0.810008	0.67	0.343372	1.17	0.098000	1.67	0.018190	2.17	0.002149	2.67	0.000159	3.17	0.00000736
0.18	0.799064	0.68	0.336218	1.18	0.095163	1.68	0.017507	2.18	0.002049	2.68	0.000151	3.18	0.00000689
0.19	0.788160	0.69	0.329160	1.19	0.092392	1.69	0.016847	2.19	0.001954	2.69	0.000142	3.19	0.00000644
0.2	0.777297	0.7	0.322199	1.2	0.089686	1.7	0.016210	2.2	0.001863	2.7	0.000134	3.2	0.00000603
0.21	0.766478	0.71	0.315335	1.21	0.087045	1.71	0.015593	2.21	0.001776	2.71	0.000127	3.21	0.00000564
0.22	0.755704	0.72	0.308567	1.22	0.084466	1.72	0.014997	2.22	0.001692	2.72	0.000120	3.22	0.00000527
0.23	0.744977	0.73	0.301896	1.23	0.081950	1.73	0.014422	2.23	0.001612	2.73	0.000113	3.23	0.00000493
0.24	0.734300	0.74	0.295322	1.24	0.079495	1.74	0.013865	2.24	0.001536	2.74	0.000107	3.24	0.00000460
0.25	0.723674	0.75	0.288845	1.25	0.077100	1.75	0.013328	2.25	0.001463	2.75	0.000101	3.25	0.00000430
0.26	0.713100	0.76	0.282463	1.26	0.074764	1.76	0.012810	2.26	0.001393	2.76	0.000095	3.26	0.00000402
0.27	0.702582	0.77	0.276179	1.27	0.072486	1.77	0.012309	2.27	0.001326	2.77	0.000090	3.27	0.00000376
0.28	0.692120	0.78	0.269990	1.28	0.070266	1.78	0.011826	2.28	0.001262	2.78	0.000084	3.28	0.00000351
0.29	0.681717	0.79	0.263897	1.29	0.068101	1.79	0.011359	2.29	0.001201	2.79	0.000080	3.29	0.00000328
0.3	0.671373	0.8	0.257899	1.3	0.065992	1.8	0.010909	2.3	0.001143	2.8	0.000075	3.3	0.00000306
0.31	0.661092	0.81	0.251997	1.31	0.063937	1.81	0.010475	2.31	0.001088	2.81	0.000071	3.31	0.00000285
0.32	0.650874	0.82	0.246189	1.32	0.061935	1.82	0.010057	2.32	0.001034	2.82	0.000067	3.32	0.00000266
0.33	0.640721	0.83	0.240476	1.33	0.059985	1.83	0.009653	2.33	0.000984	2.83	0.000063	3.33	0.00000249
0.34	0.630635	0.84	0.234857	1.34	0.058086	1.84	0.009264	2.34	0.000935	2.84	0.000059	3.34	0.00000232
0.35	0.620618	0.85	0.229332	1.35	0.056238	1.85	0.008889	2.35	0.000889	2.85	0.000056	3.35	0.00000216
0.36	0.610670	0.86	0.223900	1.36	0.054439	1.86	0.008528	2.36	0.000845	2.86	0.000052	3.36	0.00000202
0.37	0.600794	0.87	0.218560	1.37	0.052688	1.87	0.008179	2.37	0.000803	2.87	0.000049	3.37	0.00000188
0.38	0.590991	0.88	0.213313	1.38	0.050984	1.88	0.007844	2.38	0.000763	2.88	0.000046	3.38	0.00000175
0.39	0.581261	0.89	0.208157	1.39	0.049327	1.89	0.007521	2.39	0.000725	2.89	0.000044	3.39	0.00000163
0.4	0.571608	0.9	0.203092	1.4	0.047715	1.9	0.007210	2.4	0.000689	2.9	0.000041	3.4	0.00000152
0.41	0.562031	0.91	0.198117	1.41	0.046148	1.91	0.006910	2.41	0.000654	2.91	0.000039	3.41	0.00000142
0.42	0.552532	0.92	0.193232	1.42	0.044624	1.92	0.006622	2.42	0.000621	2.92	0.000036	3.42	0.00000132
0.43	0.543113	0.93	0.188437	1.43	0.043143	1.93	0.006344	2.43	0.000589	2.93	0.000034	3.43	0.00000123
0.44	0.533775	0.94	0.183729	1.44	0.041703	1.94	0.006077	2.44	0.000559	2.94	0.000032	3.44	0.00000115
0.45	0.524518	0.95	0.179109	1.45	0.040305	1.95	0.005821	2.45	0.000531	2.95	0.000030	3.45	0.00000107
0.46	0.515345	0.96	0.174576	1.46	0.038946	1.96	0.005574	2.46	0.000503	2.96	0.000028	3.46	0.00000099
0.47	0.506255	0.97	0.170130	1.47	0.037627	1.97	0.005336	2.47	0.000477	2.97	0.000027	3.47	0.00000092
0.48	0.497250	0.98	0.165769	1.48	0.036346	1.98	0.005108	2.48	0.000453	2.98	0.000025	3.48	0.00000086
0.49	0.488332	0.99	0.161492	1.49	0.035102	1.99	0.004889	2.49	0.000429	2.99	0.000024	3.49	0.00000080